

Solid state homework 1

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Problem 1

Question 1

The basis functions are defined as

$$b_j(x) = x(1-x)x^j.$$

Each basis function satisfies the boundary conditions of the infinite square well. At $x = 0$,

$$b_j(0) = 0,$$

and at $x = 1$,

$$b_j(1) = 0.$$

Therefore, any trial wavefunction of the form

$$\psi_M(x) = \sum_{j=0}^M c_j b_j(x)$$

also satisfies

$$\psi_M(0) = \psi_M(1) = 0.$$

A generalized eigenvalue problem is required because the basis functions are not orthogonal. The overlap matrix is defined by

$$S_{ij} = \langle b_i | b_j \rangle = \int_0^1 b_i(x) b_j(x) dx.$$

For example,

$$S_{01} = \int_0^1 b_0(x) b_1(x) dx = \int_0^1 x^3 (1-x)^2 dx \neq 0.$$

Thus, the basis functions are nonorthogonal and the overlap matrix is not the identity matrix,

$$S \neq I.$$

Therefore, the variational problem leads to the generalized eigenvalue equation

$$\boxed{Hc = ES c},$$

rather than the ordinary eigenvalue equation $Hc = Ec$. The matrix S accounts for the overlap between the nonorthogonal basis functions.

Question 2

The basis functions are

$$b_j(x) = x(1-x)x^j = x^{j+1}(1-x).$$

The overlap matrix elements are

$$S_{ij} = \int_0^1 b_i(x)b_j(x) dx.$$

For the Hamiltonian,

$$H_{ij} = \frac{1}{2} \int_0^1 b'_i(x)b'_j(x) dx.$$

The product rule gives

$$b'_j(x) = (j+1)x^j(1-x) - x^{j+1}.$$

Thus,

$$\boxed{b'_j(x) = (j+1)x^j - (j+2)x^{j+1}}.$$

Similarly,

$$\boxed{b'_i(x) = (i+1)x^i - (i+2)x^{i+1}}.$$

The matrix elements S_{ij} and H_{ij} were evaluated numerically for $M = 0, \dots, 5$ using 200-point Gauss-Legendre quadrature.

The generalized eigenvalue problem

$$Hc = ES c$$

was then solved numerically. The eigenvalues obtained were

M	Eigenvalues
0	5.000000
1	5.000000, 21.000000
2	4.93487481, 21.000000, 51.06512519
3	4.93487481, 19.75077641, 51.06512519, 100.24922360
4	4.93480222, 19.75077641, 44.58681182, 100.24922360, 175.47838596
5	4.93480222, 19.73923670, 44.58681182, 79.99595777, 175.47838596, 285.26480551

As M increases, the number of basis functions increases from 1 to $M + 1$. Therefore, the generalized eigenvalue problem produces $M + 1$ eigenvalues for each value of M .

The lowest eigenvalue approaches the exact ground-state energy

$$E_0 = \frac{\pi^2}{2} \approx 4.93480220$$

from above as the size of the trial space increases.

Question 3

The exact ground-state energy is

$$E_0 = \frac{\pi^2}{2} \approx 4.9348022005.$$

Using the lowest eigenvalue obtained for each M , the results are

M	E_M	$E_M - \pi^2/2$
0	5.000000000000	$6.519779945523 \times 10^{-2}$
1	5.000000000000	$6.519779945523 \times 10^{-2}$
2	4.934874810658	$7.261011367543 \times 10^{-5}$
3	4.934874810658	$7.261011367543 \times 10^{-5}$
4	4.934802217493	$1.694847018285 \times 10^{-8}$
5	4.934802217493	$1.694847018285 \times 10^{-8}$

For every value of M ,

$$E_M - \frac{\pi^2}{2} > 0,$$

so the variational energy remains above the exact ground-state energy. This demonstrates the variational upper-bound property.

The sequence of lowest energies is

$$5.000000 \rightarrow 5.000000 \rightarrow 4.93487481 \rightarrow 4.93487481 \rightarrow 4.93480222 \rightarrow 4.93480222.$$

Therefore, the sequence is monotonically nonincreasing. As M increases, the lowest energy either decreases or remains unchanged, but it does not increase. The values also approach the exact ground-state energy as the trial space becomes larger.

Question 4

For $M = 5$, the coefficient vector corresponding to the lowest eigenvalue was taken from the generalized eigenvalue problem

$$Hc = ES c.$$

Because the basis is nonorthogonal, the normalization condition is

$$c^T S c = 1.$$

Using the $M = 5$ coefficient vector, this gave

$$c^T S c = 1.0.$$

Therefore, the $M = 5$ trial wavefunction is already normalized.

The normalized $M = 5$ trial wavefunction was then constructed as

$$\psi_5(x) = \sum_{j=0}^5 c_j b_j(x).$$

It was evaluated on a grid of 1000 points and compared with the exact ground-state wavefunction

$$\psi_0(x) = \sqrt{2} \sin(\pi x).$$

The maximum absolute difference was

$$\max_x |\psi_5(x) - \psi_0(x)| = 1.4483 \times 10^{-5}.$$

The two wavefunctions look similar over the interval $0 \leq x \leq 1$, with only a small difference visible when zooming extremely closely.

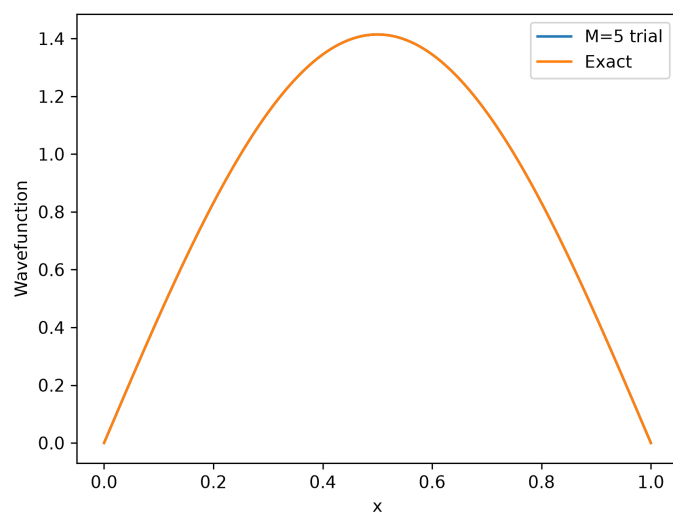


Figure 1: Comparison of the $M = 5$ trial wavefunction with the exact ground-state wavefunction.

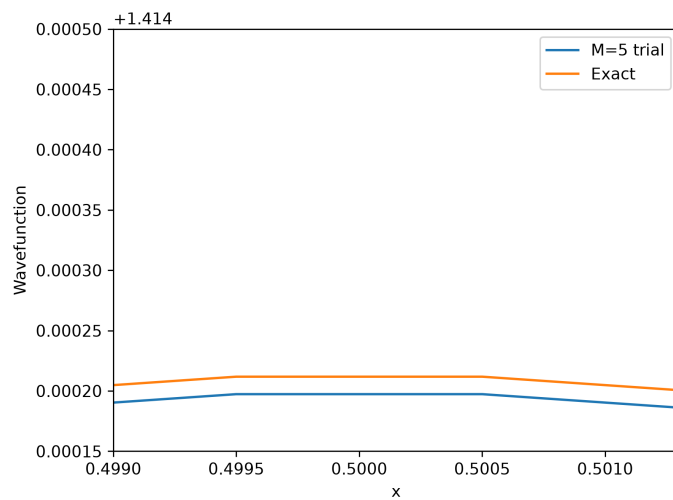


Figure 2: Zoomed view near the peak of the two wavefunctions.

Question 5

Adding a new basis function enlarges the trial space, so the variational method has more possible trial wavefunctions to choose from. Since the previous trial space is still contained in the new one, the lowest variational energy cannot increase:

$$E_{M+1} \leq E_M.$$

In this problem, the ground state is symmetric about $x = 1/2$. Every other increase in M adds a new independent antisymmetric direction. Since an antisymmetric component does not improve the approximation to the symmetric ground state, the lowest energy remains unchanged for those steps.

This explains the pattern

$$E_0 = E_1, \quad E_2 = E_3, \quad E_4 = E_5.$$

Problem 2

Question 1

All integer triples (n_x, n_y, n_z) with

$$-8 \leq n_x, n_y, n_z \leq 8$$

were enumerated in Python. For each triple,

$$q = n_x^2 + n_y^2 + n_z^2$$

was calculated, and states with the same value of q were grouped together. The orbital degeneracy $g(q)$ is the number of distinct integer triples corresponding to the same value of q . Since each spatial orbital can hold two electrons of opposite spin, the spin-inclusive shell capacity is $2g(q)$.

q	$g(q)$	$2g(q)$
0	1	2
1	6	12
2	12	24
3	8	16
4	6	12
5	24	48
6	24	48
8	12	24
9	30	60
10	24	48
11	24	48
12	8	16

Question 2

The 54 electrons were filled from the lowest-energy shell upward using the spin-inclusive shell capacity $2g(q)$.

q	$2g(q)$	cumulative electrons
0	2	2
1	12	14
2	24	38
3	16	54

Therefore, the last occupied shell is

$$\boxed{q_F = 3}.$$

The finite-cell Fermi wavevector is

$$k_F^{\text{finite}} = \frac{2\pi}{L} \sqrt{q_F}.$$

Using $L = 1.00 \times 10^{-9} \text{ m}$ and $q_F = 3$,

$$\boxed{k_F^{\text{finite}} = 1.0882796 \times 10^{10} \text{ m}^{-1}}$$

The finite-cell Fermi energy is

$$E_F^{\text{finite}} = \frac{\hbar^2 (k_F^{\text{finite}})^2}{2m_e}.$$

Using SI constants and converting the result from joules to electronvolts,

$$\boxed{E_F^{\text{finite}} = 4.51236 \text{ eV}}$$

Question 3

The electron number density is

$$n = \frac{N}{L^3}.$$

Using $N = 54$ and $L = 1.00 \times 10^{-9} \text{ m}$,

$$\boxed{n = 5.40 \times 10^{28} \text{ m}^{-3}}$$

The continuum Fermi wavevector is

$$k_F^{\text{cont}} = (3\pi^2 n)^{1/3},$$

which gives

$$\boxed{k_F^{\text{cont}} = 1.16933 \times 10^{10} \text{ m}^{-1}}$$

The corresponding continuum Fermi energy is

$$E_F^{\text{cont}} = \frac{\hbar^2 (k_F^{\text{cont}})^2}{2m_e},$$

giving

$$E_F^{\text{cont}} = 5.20954 \text{ eV}$$

The percent difference from the finite-cell value is

$$\frac{|E_F^{\text{cont}} - E_F^{\text{finite}}|}{E_F^{\text{finite}}} \times 100 = 15.45\%.$$

The difference is because in the finite cell, the allowed k -states are discrete, while the continuum approximation treats the allowed states as continuously distributed in k -space.

Question 4

The shell capacity $2g(q)$ for the first twelve allowed shells was plotted as a stem plot.

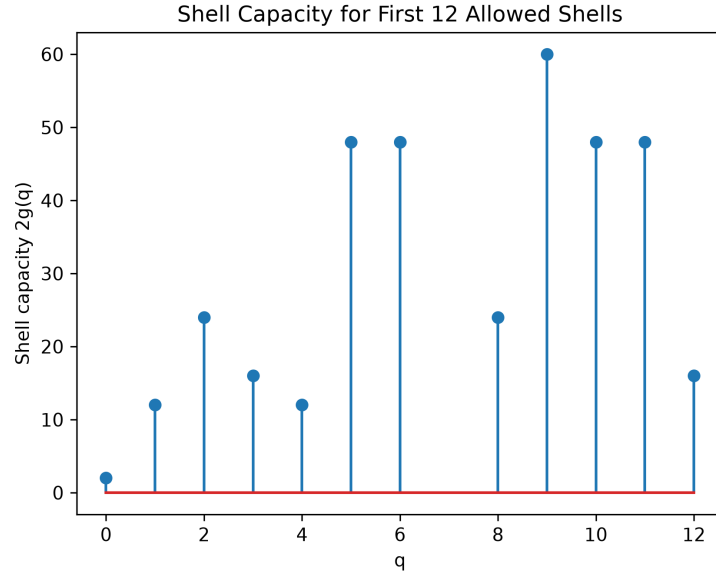


Figure 3: Spin-inclusive shell capacity $2g(q)$ for the first twelve allowed shells.

The discrete density of states was also plotted using the first twelve allowed shell energies,

$$E_q = E_0 q,$$

with the number of states in each shell given by $2g(q)$. The finite-cell and continuum Fermi energies are marked for comparison.

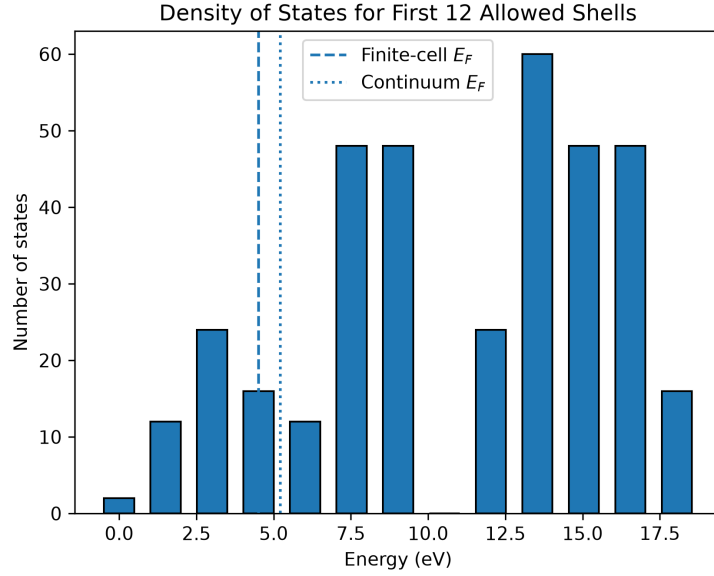


Figure 4: Discrete density of states for the first twelve allowed shells. The dashed line marks the finite-cell Fermi energy and the dotted line marks the continuum estimate.

Question 5

For a given shell,

$$q = n_x^2 + n_y^2 + n_z^2,$$

the orbital degeneracy $g(q)$ is the number of distinct integer triples (n_x, n_y, n_z) that produce the same value of q .

The $+\mathbf{k}$ and $-\mathbf{k}$ states are already counted as different spatial states whenever they correspond to different integer triples. For example,

$$(1, 0, 0)$$

and

$$(-1, 0, 0)$$

are two distinct spatial orbitals and are both included in $g(q)$.

Each spatial orbital can then hold two electrons with opposite spin, so the spin-inclusive shell capacity is

$$2g(q).$$

Therefore, orbital degeneracy and the $+\mathbf{k}/-\mathbf{k}$ distinction are included when counting the distinct integer triples, while spin adds one additional factor of 2. None of these factors should be inserted twice, because that would double-count the same quantum states.

Problem 3

Question 1

First,

$$F(x) = 1 + \frac{1-x^2}{2x} \ln \left| \frac{1+x}{1-x} \right|.$$

The direct expression is undefined at $x = 0$ and $x = 1$, but both points are removable singularities. Using the analytic limits given in the problem,

$$F(0) = 2, \quad F(1) = 1.$$

These removable singularities were handled directly in the Python function. When $x = 0$, the function returns the analytic limit

$$F(0) = 2.$$

When $x = 1$, it returns the analytic limit

$$F(1) = 1.$$

For all other values of x , the original expression for $F(x)$ is evaluated directly. As a check, the implementation gives

$$F(0) = 2.0, \quad F(1) = 1.0, \quad F(0.5) = 1.8239592165.$$

Question 2

For $r_s = 1$ and $r_s = 2$, the free-electron dispersion

$$\epsilon_{\text{free}}(k) = \frac{k^2}{2}$$

and the Hartree–Fock dispersion

$$\epsilon_{\text{HF}}(k) = \frac{k^2}{2} + \Sigma_x(k)$$

were plotted over

$$0 \leq \frac{k}{k_F} \leq 1.5.$$

The occupied region $0 \leq k/k_F \leq 1$ is shaded, and the dashed line marks $k = k_F$.

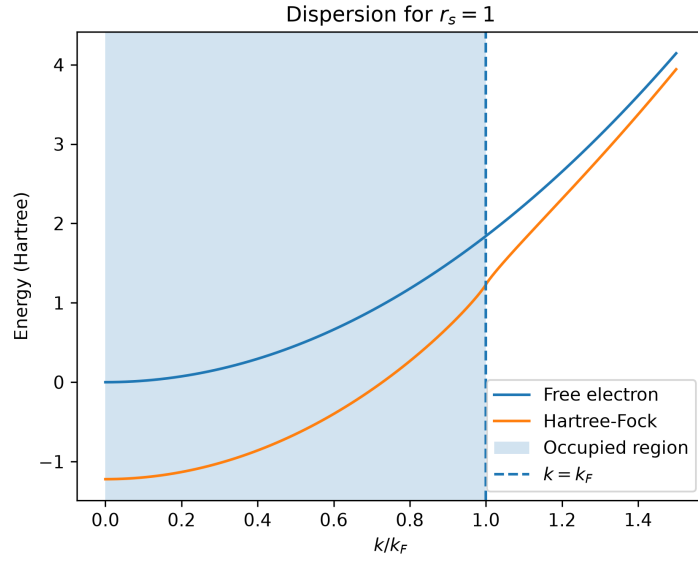


Figure 5: Free-electron and Hartree-Fock dispersions for $r_s = 1$.

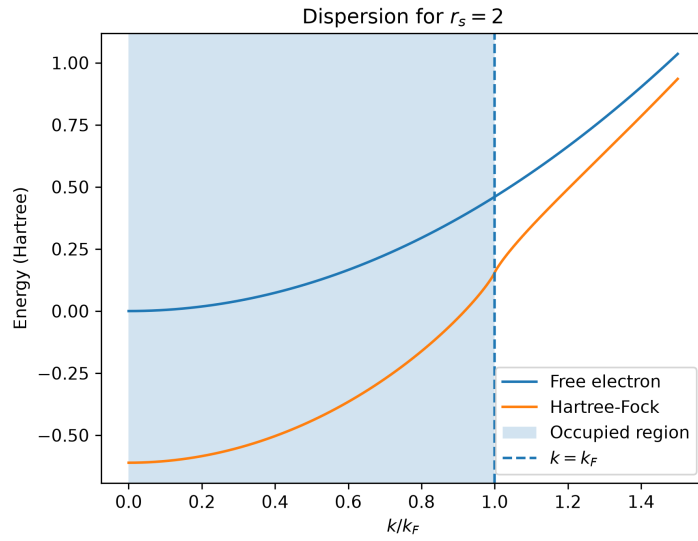


Figure 6: Free-electron and Hartree-Fock dispersions for $r_s = 2$.

Question 3

For each value of r_s , the Fermi wavevector was calculated using

$$k_F = \frac{1}{r_s} \left(\frac{9\pi}{4} \right)^{1/3}.$$

The energies were then evaluated from

$$\epsilon_{\text{free}}(k_F) = \frac{k_F^2}{2},$$

$$\epsilon_{\text{HF}}(k_F) = \frac{k_F^2}{2} - \frac{k_F}{\pi},$$

$$\frac{T}{N} = \frac{3k_F^2}{10},$$

$$\frac{E_x}{N} = -\frac{3k_F}{4\pi},$$

and

$$\frac{E_{\text{HF}}}{N} = \frac{3k_F^2}{10} - \frac{3k_F}{4\pi}.$$

The energies were converted from Hartree to electron volts using

$$1 \text{ Ha} = 27.2114 \text{ eV}.$$

r_s	$k_F \text{ (}\text{\AA}^{-1}\text{)}$	$\epsilon_{\text{free}}(k_F)$	$\epsilon_{\text{HF}}(k_F)$	T/N	E_x/N	E_{HF}/N
1	1.91916	50.1121 eV	33.4890 eV	30.0673 eV	-12.4673 eV	17.5999 eV
2	0.959579	12.5280 eV	4.21648 eV	7.51681 eV	-6.23366 eV	1.28315 eV

The quantity $\epsilon_{\text{HF}}(k_F)$ is the Hartree–Fock energy of a single orbital located at the Fermi surface. In contrast, E_{HF}/N is the total Hartree–Fock energy of the entire electron gas divided by the number of electrons.

For $r_s = 1$,

$$\epsilon_{\text{HF}}(k_F) = 33.4890 \text{ eV}, \quad \frac{E_{\text{HF}}}{N} = 17.5999 \text{ eV},$$

while for $r_s = 2$,

$$\epsilon_{\text{HF}}(k_F) = 4.21648 \text{ eV}, \quad \frac{E_{\text{HF}}}{N} = 1.28315 \text{ eV}.$$

These values are different because $\epsilon_{\text{HF}}(k_F)$ is for one orbital at $k = k_F$, but E_{HF}/N contains the average kinetic and exchange contributions of all occupied electrons in the Fermi sea.

Question 4

For $r_s = 2$, the derivative of the Hartree–Fock dispersion at the Fermi surface was estimated using the symmetric finite-difference formula

$$\left. \frac{d\epsilon_{\text{HF}}}{dx} \right|_{x=1} \approx \frac{\epsilon_{\text{HF}}(1+\delta) - \epsilon_{\text{HF}}(1-\delta)}{2\delta}.$$

The resulting values are

δ	$d\epsilon_{\text{HF}}/dx _{x=1}$ (Ha)
10^{-1}	1.839284
10^{-2}	2.539198
10^{-3}	3.242440
10^{-4}	3.945748
10^{-5}	4.649058
10^{-6}	5.352368

As δ decreases, the estimated slope continues to increase rather than approaching a finite value. This indicates that the Hartree–Fock dispersion becomes singularly steep at the Fermi surface.

Since the density of states satisfies

$$g(E) \propto \frac{k^2}{dE/dk},$$

the increasing slope implies that the Hartree–Fock density of states decreases toward zero at the Fermi energy,

$$g(E_F) \rightarrow 0.$$

Question 5

Exchange acts only between electrons with the same spin because spin-up and spin-down states are orthogonal. Therefore, the exchange term vanishes for opposite-spin electrons.

Exchange is nonlocal because its value at one position depends on the electron orbital at other positions in space, not only at that same point.

Even though exchange is nonlocal, plane waves remain eigenfunctions in a homogeneous electron gas. The system is translationally invariant, so applying the exchange operator to a plane wave gives back the same plane wave multiplied by a k -dependent factor. Therefore, the exchange interaction changes the energy of the plane-wave state without changing its form.

Python Source Code

The documented Python source files for each problem are submitted separately.